

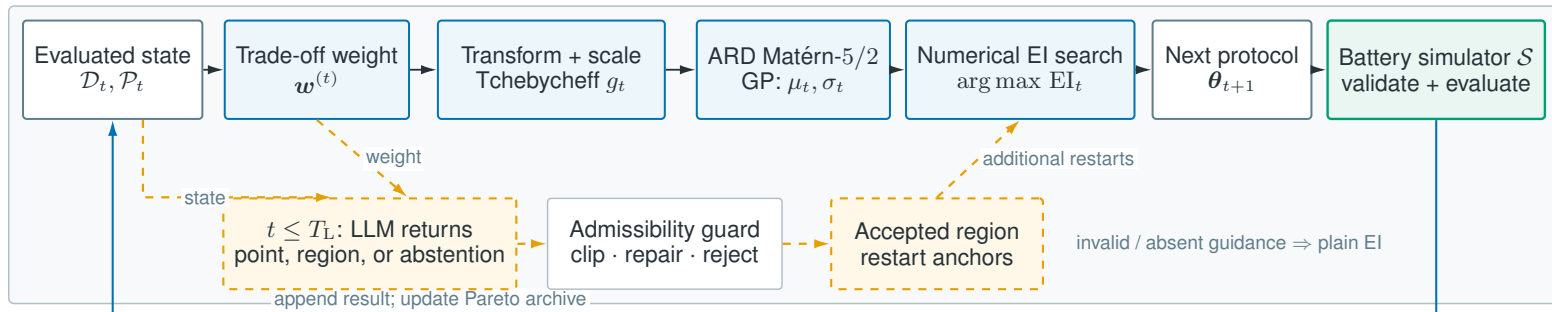
(a) Simulator-validated warm start

Global prior before sequential Bayesian optimization



(b) Sequential decomposition-based MOBO

Numerical backbone remains active for the full evaluation budget



Validation firewall: the LLM proposes search locations only. Only simulator-evaluated protocols determine objectives, feasibility, GP targets, and Pareto updates.

